EXPERIMENT- 09

Python Code:

09.py > ...

```python
keyword = input("Enter keyword: ").upper().replace("J", "I").replace(" ", "")
ciphertext = input("Enter ciphertext: ").upper().replace("J", "I").replace(" ", "")

matrix = []
used = set()
for char in keyword + "ABCDEFGHIKLMNOPQRSTUVWXYZ":
    if char not in used:
        matrix.append(char)
        used.add(char)
matrix = [matrix[i:i+5] for i in range(0, 25, 5)]

def get_pos(c):
    for i in range(5):
        for j in range(5):
            if matrix[i][j] == c:
                return i, j

decrypted_text = ""
for i in range(0, len(ciphertext) - 1, 2):
    r1, c1 = get_pos(ciphertext[i])
    r2, c2 = get_pos(ciphertext[i+1])

    if r1 == r2:
        decrypted_text += matrix[r1][(c1-1)%5] + matrix[r2][(c2-1)%5]
    elif c1 == c2:
        decrypted_text += matrix[(r1-1)%5][c1] + matrix[(r2-1)%5][c2]
    else:
        decrypted_text += matrix[r1][c2] + matrix[r2][c1]

print("Decrypted Text:", decrypted_text)
```

Output:

```
PS D:\College\Crypto\Experiments> python -u "d:\College\Crypto\Experiments\09.py"
Enter keyword: ROYAL NEW ZEALAND NAVY
Enter ciphertext: KXJEY UREBE ZWEHE WRYTU HEYFS
Decrypted Text: PTBOATONEOWENINELOSTINAC
PS D:\College\Crypto\Experiments>
```